

are Photoluminescent Quantum Dots (PL-QDs) which emit light when excited by UV light or blue light.

QLED Displays

QLEDs are the most popular type of Quantum Dots-based displays in consumer devices today. These displays use a quantum dot enhancement layer (either QDEF film or xQDEF diffuser plate) between the backlight and LCD panel.

The blue light from the backlight LEDs falls on Quantum Dots and some of it is converted to Green and Red. This results in a healthy mix of Red, Green and Blue components in the white light which is necessary to achieve a larger colour volume.

QLED TVs are easy to manufacture which is also why they are quite popular. LG's QNED TVs and some of Sony's Triluminous TVs are also based on the same technology. The MacBook Pro launched in 2024 also uses QDEF film in its display for the first time as was discovered by DSCC.

QD-OLED Displays

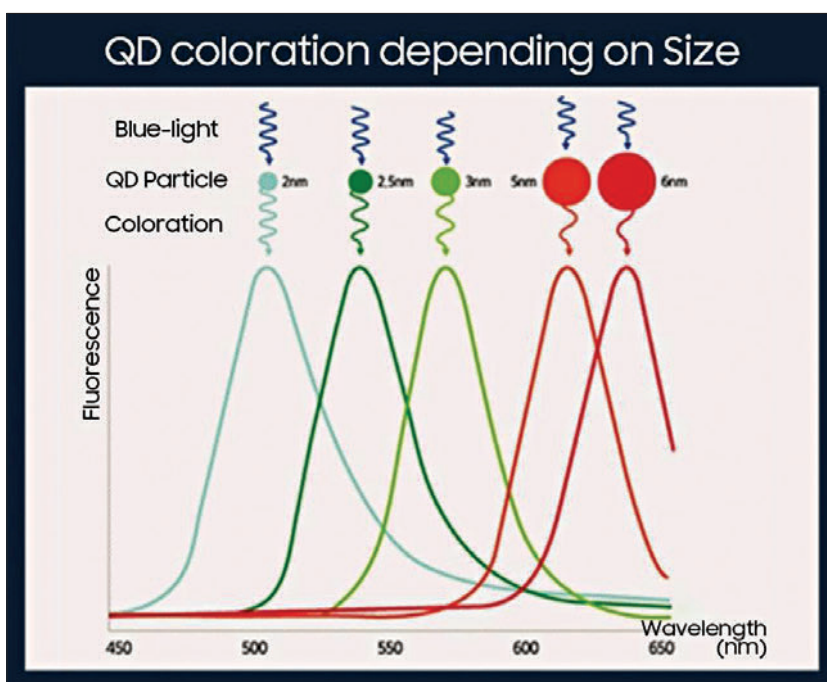
QD-OLED uses a Quantum Dots layer for colour conversion, unlike WHITE OLEDs which use a colour filter. It's easy to manufacture OLED displays with red, blue and green subpixels for smartphones but not for larger display sizes like TVs and monitors.

Instead, manufacturers use a single colour emitter and then achieve colour conversion using a colour filter or quantum dots. When Quantum Dots are used, there is no loss in quantum efficiency and the display can achieve larger colour volume and higher contrast. This also enables QD-OLED TVs to be brighter than conventional white OLED TVs.

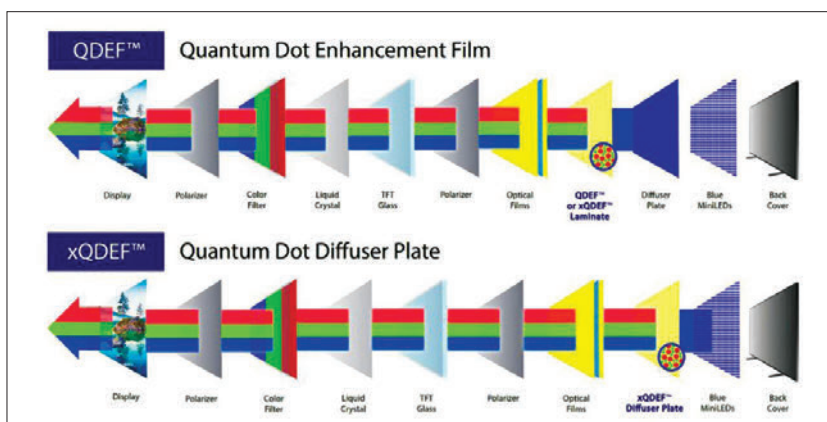
QD-micro LED Displays

QD-micro LED displays are principally similar to the QD-OLED displays. The difference is that instead of organic emitters or OLEDs, these displays use inorganic emitters or micro LEDs which can last longer and can go much brighter.

Again, it's challenging to transfer three different colour micro LEDs to



Source: Samsung



Source: Nanosys

a substrate and thus manufacturers see using a single blue colour emitter followed by a Quantum Dot colour conversion layer in later stages as a more pragmatic approach. QD-micro LED displays are still a few years away from consumer devices.

QDEL or QD-LED or Nano LED Display

QDEL is one of the most promising upcoming display technologies. It is also referred to as Nano LED or QD-LED. In fact, there were times when this technology was also referred to as QNED and QLED, which were later registered by LG and Samsung for different display technologies and are now quite popular.

All the other applications that we have discussed above use Photoluminescent Quantum Dots that are excited using blue light. QD-EL or Nano LEDs on the other hand use Electroluminescent Quantum Dots which like regular LEDs can emit light when excited by current. In QDEL displays, Quantum dots do everything on their own. They don't require an OLED or LED layer and thus offer high purity and brightness.

The technology was the prime focus at SID Display Week and TCL also showcased a 14-inch 2.8K QD-EL laptop screen prototype fabricated by ink-jet printing. This display offers a variable refresh rate (30 Hz – 120 Hz) and a wide viewing angle. 